

EduCredChain, A Hybrid Blockchain-PKI Architecture for Educational Credential Validation.

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Abstract - Secure digital credential management requires trustworthy and tamper-resistant mechanisms to prevent certificate forgery and replace inefficient manual validation. This paper proposes a hybrid framework with Blockchain and PKI for securely issuing, verifying, and revoking educational credentials. EduCredChain employs standardized cryptographic algorithms (RSA, AES-256, SHA-256) and hybrid storage with on-chain and off-chain data storage model. to support interoperability and secure credential lifecycle management framework, EduCredChain combines blockchain immutability with public key infrastructure (PKI) to create trust while keeping privacy intact. Certificate data is stored off-chain storage while certificate hash and revocation details are saved on the blockchain. which reduce gas cost, storage needs and limits data exposure. RSA digital signatures ensure issuer and certificate authenticity, integrity, and non-repudiation. AES-256 encryption protects issuer private key, which is decrypted only during certificate signing. Smart contracts handle the registration, verification, and revocation of certificates automatically, allowing transparent and tamper-resistant credential management throughout the credential lifecycle. A prototype was created using python, Ethereum based ganache blockchain, Solidity, and SQLite. Experimental evaluation demonstrates successful operational features, low verification latency, and reduced blockchain gas consumption. This framework delivers a practical and scalable method to secure and reliably manage digital credentials. This proposed framework has the potential to serve as a generic secure document verification platform beyond the education sector.

Keywords: *Blockchain; Educational Credential Verification; Public Key Infrastructure (PKI); Smart Contracts; Cryptography; Key Management; Certificate Revocation; SHA-256 Hashing; Secure Credential Management; RSA signature; Security Architecture; Hybrid Storage;*

1. Introduction

1.1 Background of Educational Credential Fraud

Educational credentials play a critical role in academic admissions, employment opportunities, professional licensing, and career advancement [1][10]. Universities, colleges, and training institutions issue certificates to certify academic achievements and qualifications[10]. As organizations depend on these credentials for decision-making, ensuring their authenticity and integrity has become a challenge. The widespread adoption of digital technologies has simplified the creation, distribution, and storage of academic certificates, also increased the risk of credential forgery[10]. Fraudulent activities such as certificate forgery, unauthorized modification of academic records, falsification of grades, and the creation of counterfeit degrees have become increasingly prevalent[2][14]. Advanced editing and digital document manipulation technologies have made it easier for malicious actors to produce convincing counterfeit certificates that are difficult to distinguish from genuine credentials. The consequences of educational credential fraud extend beyond individual institutions and affect employers, governments, and society as a whole. Fraudulent qualifications enable unqualified individuals obtaining employment, professional certifications, or academic admissions, thereby undermining trust in educational systems and professional standards. These challenges highlight the need for secure, tamper-resistant, and efficient credential verification mechanisms capable of ensuring authenticity while supporting scalable digital credential management.

1.2 Limitations of Traditional Methods

Traditional educational credential verification systems mostly rely on centralized databases, paper-based physical records, and manual verification procedures[20][21]. While these methods have been widely used for decades, they suffer from many limitations that

affect the security, efficiency, and reliability of credentials[18]. One of the major challenges is the dependence on a centralized authority[25][27], where all credential records are maintained within institutional databases. Such systems are vulnerable to single points of failure, unauthorized access, data breaches, and accidental data loss. Another significant limitation is the chance of physical certificate forgery and document tampering. Certificates can be easily duplicated, altered, or fabricated using various modern tools[32]. Detecting modifications requires direct verification from the issuer. In many cases, this is done by manually contacting universities or training organizations to confirm the authenticity of credentials which causes delay and increased administrative overhead[18][21]. These are some of the challenges faced by traditional verification mechanisms. In addition to that, modern cyberattacks such as database poisoning, unauthorized record manipulation, insider threats, and targeted database attacks can alter or inject fraudulent records into databases thereby compromising the integrity of credentials. This causes verification systems to produce false-positive results, leading to the acceptance of forged or invalid credentials as legitimate[5][28]. Existing systems often provide limited support for secure credential revocation and lifecycle management[8][31]. Scalability is also a concern, particularly as educational institutions continue to generate large volumes of digital credentials. Managing, verifying, and maintaining these records through conventional methods has become more complex[3][25].

1.3 Role of blockchain

Blockchain technology provides a robust framework to securely record and verify digital transactions[27][36]. Unlike traditional centralized databases, blockchain stores data across multiple nodes in a distributed network, eliminating dependence on a single server which reduces the risk of unauthorized tampering[6][31]. Once data is recorded and validated on the blockchain, it becomes extremely difficult to alter or delete, ensuring long-term integrity and transparency which is ideal for credential storage[3][21][28]. Through the use of cryptographic hashing and consensus mechanisms, blockchain ensures that stored records are tamper-resistant and can be verified by any authorized participants. Any attempts to modify blocks will result in inconsistencies that can be readily detected[12][28]. In recent years, blockchain has been adopted across various domains, including finance, healthcare, supply chain management, digital identity systems, and educational credential management[26][38]. By providing a transparent and auditable record of

transactions, blockchain enables stakeholders to independently verify the authenticity of stored information while maintaining a high level of security and reliability[8][33].

1.4 Role of Public Key Infrastructure

Public Key Infrastructure (PKI) is a widely adopted security framework that enables secure communication, authentication, and data protection by implementing asymmetric cryptography[28][31]. The operation of PKI is closely related to the concept of digital signatures which establishes trust among entities by managing digital certificates and cryptographic keys, allowing users and systems to verify and ensure identity and authenticity of stored credentials[4][6]. It acts as a fundamental component supporting applications such as secure web communications, digital signatures, encrypted messaging, and identity management. PKI will provide reliable authentication through digital signatures[28]. Each entity is assigned a pair of cryptographic keys consisting of a public key and a private key. The private key which is used to generate digital signatures is kept confidential, while the corresponding public key which is used to verify signatures is open to all. This mechanism ensures that the certificate is issued by an authorized party who has access to the secret private key thereby strengthening security and trust. Later anyone possessing the corresponding public key can verify that the data was generated by the legitimate authority and not altered since signing[28]. This capability provides non-repudiation, ensuring that the signer cannot later deny their involvement in the transaction. By providing authentication, integrity, and non-repudiation, PKI forms a reliable trust framework for protecting sensitive digital assets[6][28]. Its established security model and widespread adoption make it an essential component in the design of secure and trustworthy credential verification systems.

1.5 Requirement of a Hybrid Architecture

The limitations of existing centralized and blockchain-based credential verification systems highlight the need for a hybrid architecture that combines the strengths of both approaches while mitigating their individual drawbacks. Centralized systems provide efficient data management and straightforward access control but remain vulnerable to database tampering or single points of failure. blockchain addresses this limitation with immutability and transparency but blockchain cannot store complete credential information on-chain transparently. Because doing so causes privacy concerns, storage

overhead, and transaction costs. So, an off-chain storage mechanism is essential but it is vulnerable to various attacks as mentioned earlier. Hence PKI is implemented with off-chain storage for authentication, digital signatures, and secure key management. Using PKI along with a database does not inherently provide a tamper-resistant record storage mechanism. Consequently, a hybrid architecture which uses both blockchain and PKI is required to address these challenges by separating sensitive credential data from public blockchain storage, while maintaining data integrity with signed credential data. This approach preserves the authenticity and integrity along with an immutable proof of credential.

1.6 Research objectives

The proposed framework aims to address the limitations of the traditional credential verification systems like certificate forgery, unauthorized modifications, inefficient verification processes and centralized trust dependencies by implementing a trusted, tamper-resistant and scalable solution for secure verification of educational credentials in academic, professional and organizational settings. The specific objectives of this research are as follows:

1. To design a hybrid Blockchain - PKI architecture that combines the immutability of blockchain technology with the authentication and trust mechanisms provided by Public Key Infrastructure.
2. To provide secure credential storage by implementing a hybrid storage model in which certificate data is maintained in signed off-chain storage while SHA-256 cryptographic hashes are preserved on the blockchain as cryptographic proof to prevent exposure of sensitive credential data on blockchain.
3. To implement PKI using RSA cryptographic operations and AES-256 encryption for secure storage and protection of the RSA private key from unauthorized access.
4. To develop a system with adequate certificate lifecycle management framework that supports the users in issuing, verifying, and revoking credentials across both on-chain and off-chain simultaneously.
5. To evaluate the security and performance of the proposed framework with respect to on-chain confidentiality, integrity, authentication, non-repudiation, and resistance against credential forgery and data tampering attacks.

1.7 Paper contributions

The major contributions of this paper are summarized as follows:

1. **A Hybrid Blockchain - PKI Architecture for Educational Credential Verification:** This paper proposes a secure distributed architecture for educational credential verification, a hybrid framework that integrates blockchain technology with Public Key Infrastructure (PKI) to establish a secure and trustworthy credential verification ecosystem. The architecture combines decentralized integrity verification with cryptographic authentication mechanisms to strengthen credential security.
2. **A Privacy Preserving Hybrid Storage Model:** The proposed framework adopts a hybrid storage model where stores data both on-chain and off-chain in which only cryptographic certificate hashes and revocation information are stored on the blockchain, while sensitive credentials are maintained in signed off-chain storage. This design reduces blockchain transaction cost while preserving data privacy.
3. **Secure Key Management Using AES-Protected RSA Cryptography:** The framework employs AES-256 encryption to safeguard issuer's RSA private key. The AES key is derived when the issuer enters issuer code hence making it impossible to steal AES key digitally because it is never stored.
4. **Integrated Credential Lifecycle Management:** EduCredChain provides a complete credential management workflow encompassing certificate issuance, verification, and revocation. Smart contracts are deployed to automate these processes and maintain a transparent and tamper-resistant record of credential status changes.
5. **Security Analysis Against Credential-Based Attacks:** The proposed system is implemented with integrity, authentication, and non-repudiation. Furthermore, its resilience against certificate forgery, unauthorized modification, database tampering, and credential manipulation attacks is evaluated.
6. **Standardized hybrid credential architecture supporting interoperable APIs and modular integration.:** A prototype implementation based on Ethereum smart contracts, Solidity, Web3.py, and SQLite is developed to demonstrate the practical feasibility of the proposed framework. Various performance and security evaluations are conducted to assess the effectiveness of the architecture in real-world credential verification scenarios.

1.8 Alignment With SDGs

- SDG 4 – Quality Education
Target 4.3: By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university.
Target 4.4: By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship.
- SDG 9 – Industry, Innovation and Infrastructure
Target 9.1: Develop quality, reliable, sustainable and resilient infrastructure, including regional and transborder infrastructure, to support economic development and human well-being, with a focus on affordable and equitable access for all.
Target 9.5: Enhance scientific research, upgrade the technological capabilities of industrial sectors in all countries, in particular developing countries, including, by 2030, encouraging innovation and substantially increasing the number of research and development workers per 1 million people and public and private research and development spending.
- SDG 16 – Peace, Justice and Strong Institutions
Target 16.6: Develop effective, accountable and transparent institutions at all levels.
Target 16.10: Ensure public access to information and protect fundamental freedoms, in accordance with national legislation and international agreements.

2. Existing Systems and Related Works

2.1 Existing Systems

2.1.1 Blockcerts [2][3][27]

Blockcerts is an open-source standard for issuing, sharing, and verifying blockchain-based digital credentials, originally developed by the MIT Media Lab in collaboration with Learning Machine[2][21]. Blockcerts utilizes blockchain technology, cryptographic hashing, digital signatures, and public key cryptography to issue and verify decentralized digital credentials. The architecture can assure decentralized, tamper-evident credential verification with learner-owned digital certificates, enabling independent verification without continuous reliance on the issuing institution[41]. However, it primarily focuses on credential issuance and verification, with limited support for a

comprehensive framework that combines decentralized credential verification with privacy-preserving hybrid storage, integrated cryptographic key protection, and efficient certificate lifecycle management[41].

2.1.2 OpenCerts [19][31][33]

OpenCerts is a blockchain-based digital credential verification platform developed by the Government Technology Agency of Singapore (GovTech) to enable secure issuance and verification of academic certificates and professional credentials. The System employ blockchain technology, cryptographic hashing, digital signatures, and decentralized verification mechanisms to authenticate educational credentials while preserving data integrity. OpenCerts effectively offers secure, blockchain-based verification of educational credentials with improved transparency, integrity, and efficient authentication through cryptographic proof. Even then OpenCerts fails to provides secure blockchain-based credential verification but offers limited flexibility for customized cryptographic key management, hybrid storage architectures, and institution-specific deployment requirements. The opencerts Current implementations leave opportunities for research into flexible hybrid Blockchain-PKI architectures with enhanced key management, off-chain storage, and institution-independent deployment models.

2.1.3 DigiLocker [20]

DigiLocker is a widely adopted digital document storage and verification platform launched by the Government of India under the Digital India initiative, to enable citizens to easily and securely access and share electronic versions of official documents, educational certificates, identity records, and government-issued credentials. This system uses a centralized cloud-based architecture integrated with Public Key Infrastructure (PKI), digital signatures, secure APIs, and encrypted document storage to issue, store, and verify digital documents. DigiLocker delivers a government-backed seamless verification of digital documents across multiple services. It depends on a centralized infrastructure, lacking the immutable, decentralized verification and tamper-resistant record management provided by blockchain-based systems. However, DigiLocker does not address decentralized credential verification frameworks that provide immutable record storage, cryptographic integrity validation, and reduced dependence on centralized trust

		<i>Blockcerts</i>	<i>Opencerts</i>	<i>Digilocker</i>	<i>EduCredChain</i>
1.	Blockchain	Yes	Yes	No	Yes
2.	PKI	Partial	Partial	Yes	Yes
3.	Hybrid Storage	Partial	Partial	No	Yes
4.	Database Tampering Detection	No	No	Partial	Yes
5.	Secure RSA Key Protection (AES-256)	No	No	No	Yes
6.	User-friendly web-based interface	Partial	Yes	Yes	Yes
7.	Blockchain Wallet Required	Yes	No	No	No
8.	Certificate Revocation	Partial	Partial	Yes	Yes

Table 1: Feature Comparison of Existing Credential Verification Systems

(Note: The comparison is based on the publicly documented features and architectural characteristics of each system as described in the cited literature)

2.2 Recent blockchain-based credential verification papers

Recent studies have explored a variety of blockchain architectures to improve credential authenticity, integrity, and verification efficiency while minimizing dependence on centralized authorities[1].

Work/Author	Technology used				
	Blockchain	Smart Contracts	PKI	Sensitive Data Stored On-Chain	Database
Said et al. [1]	BSC	Yes	No	Yes	No
Kabashi et al. [4]	Ethereum	Yes	No	No	No
Leka et al. [13]	Ethereum	Yes	No	Yes	Yes
Mohan et al. [32]	Ethereum	No	No	No	No
Raju et al. [39]	Ethereum	Yes	No	No	No
UNIC [42]	Bitcoin	-	Partial	No	No
Gräther et al. [31]	Ethereum	Yes	No	No	No
EduCTX [3]	ARK	No	No	No	No
BCert [7]	Ethereum	Yes	No	No	No
Gresch et al. [6]	Ethereum	No	No	No	No
Cerberus [8]	Ethereum	No	No	Yes	Yes
VECefblock [9]	Hyperledger	No	No	No	No
Hsu et al. [10]	Hyperledger	No	No	Yes	Yes
Educ-Dapp [33]	Ethereum	No	No	No	Yes
Palma et al. [44]	Ethereum	No	No	No	Yes
Xie et al. [46]	Ethereum	No	No	No	No
Vidal et al. [47]	Ethereum	No	No	No	Yes
DCVPC [43]	Hyperledger	No	No	Yes	No
CryptoCert [48]	Ethereum	No	No	No	No
Serranito [45]	Ethereum	Yes	No	Yes	Yes
EduCredChain (Proposed System)	Ethereum	Yes	Yes	No	Yes

Table 2: Comparison of Recent Blockchain-Based Educational Credential Verification Systems

2.3 Research gaps

The analysis of existing credential verification systems and recent blockchain-based researches reveals the significant challenges in credential management systems. The existing platforms such as Blockcerts and OpenCerts successfully leverage blockchain technology while they offer limited support for modern storage architectures, comprehensive cryptographic key management, and privacy-preserving protection of sensitive credential data. Conversely, DigiLocker provides secure document management through a centralized PKI-based infrastructure but lacks in decentralized trust and immutable record storage capabilities inherent to blockchain systems. Furthermore, a review of recent blockchain-based credential verification studies indicates that most proposed solutions primarily emphasize blockchain implementation and smart contracts while providing limited integration of PKI, key protection, off-chain data storage, and comprehensive certificate lifecycle management. Consequently, there remains a need for a hybrid Blockchain–PKI framework that combines decentralized integrity verification with secure key management, off-chain storage, user-friendly verification, and efficient certificate issuance, verification, and revocation. Addressing these research gaps is the primary motivation for the proposed EduCredChain architecture, which aims to provide a secure, efficient, and privacy-preserving hybrid Blockchain–PKI architecture for educational credential verification.

3. System Architecture

3.1 System Overview

EduCredChain is a hybrid Blockchain–PKI framework integrates blockchain technology, PKI, cryptographic hashing, and RSA based signature to ensure the authenticity, integrity, and validity of digital certificates on blockchain throughout their lifecycle. Unlike conventional systems, EduCredChain framework adopts a hybrid storage strategy in which sensitive certificate information is stored in signed off-chain storage, whereas only cryptographic proofs and certificate status information are recorded on the

blockchain. This design minimizes blockchain storage requirements and transaction cost while enabling independent verification of credential integrity through immutable blockchain records. Furthermore, the integration of PKI provides trusted authentication ensuring that only authorized institutions can issue and manage credential and act as a digital signature. EduCredChain supports the complete credential lifecycle, including certificate issuance, verification, and revocation. By combining decentralized trust, cryptographic security, and privacy-preserving storage, EduCredChain provides a scalable and tamper-resistant solution for modern educational credential management.

3.2 Overall Architecture

The proposed EduCredChain architecture (fig 1) follows a layered hybrid Blockchain–PKI model comprising three major components: the User Interface, Backend Layer, and Data Storage Layer. Users interact with the system through a web-based interface to perform certificate issuance, verification, and revocation requests. These requests are processed by the backend server, which acts as the central controller coordinating cryptographic operations, blockchain interactions, and database management. Within the backend, a SHA-256 hash generator computes a unique cryptographic fingerprint for each certificate to ensure integrity. Blockchain operations communicate with the Ethereum smart contract to record the certificate hash, timestamp, and revocation status on the blockchain, thereby providing tamper-resistant verification. Simultaneously, database operations securely store the certificate data in an SQL database after signing with RSA-based signature. To enhance key security, the RSA private key is protected using AES256 encryption and is decrypted only when cryptographic operations such as digital signature generation are required. Certificate verification is performed using the RSA verification engine with the corresponding public key, while integrity is validated by comparing the computed certificate hash against the immutable blockchain record. This hybrid architecture combines blockchain immutability, PKI-based authentication, and signed off-chain storage to provide secure, privacy-preserving, and efficient educational credential management.

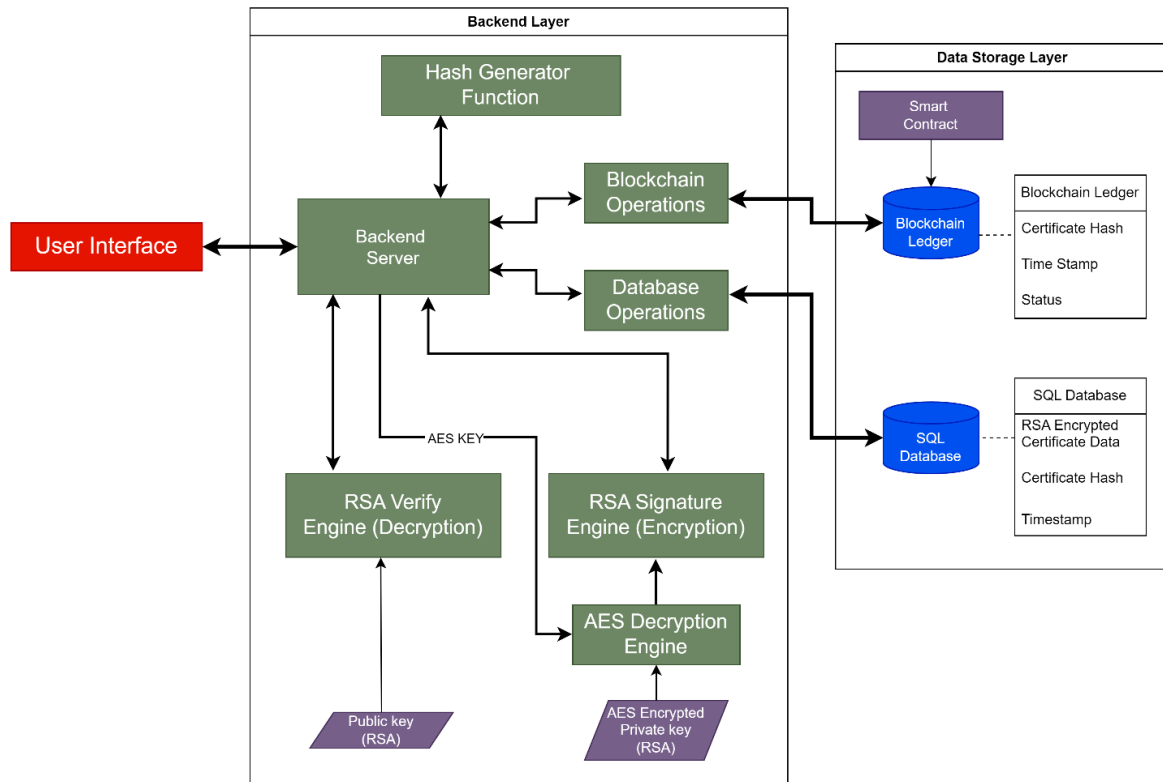


Fig 1, Architecture diagram of EduCredChain framework

3.3 System Components

3.3.1 User Interface,

The User Interface (UI) provides a web-based platform through which issuers and verifiers can directly interact with the EduCredChain system. It provides a unified, simple, and user-friendly platform for certificate issuance, verification, and revocation.

3.3.2 Backend Server,

The Backend Server acts as the central controller of the system, coordinating various internal operations along with blockchain interactions, and database transactions while processing user requests securely and efficiently. The Backend Server acts as an intermediary between the user interface and the hybrid storage, securely coordinating system operations.

3.3.3 Hash Generation Module,

The Hash Generation Module is responsible for computing a unique SHA-256 cryptographic hash for each certificate from the credential details. The generated SHA-256 hash serves as a tamper-evident digital fingerprint of the certificate and is stored on the blockchain instead of the raw credential data, thereby preserving data privacy while enabling integrity verification.

3.3.4 RSA Signature Engine,

The RSA Signature Engine digitally signs certificate data using the issuer's RSA private key before it is stored in the SQL database. The resulting digital signature enables subsequent verification of the certificate's authenticity and integrity using the corresponding public key.

3.3.5 RSA Verification Engine,

The RSA Verification Engine verifies the digital signature using the issuer's RSA public key during certificate verification. Successful verification confirms that the certificate was issued by the legitimate issuer and that its contents have not been modified, thereby ensuring authenticity, integrity, and non-repudiation.

3.3.6 AES Decryption Engine,

The AES Decryption Engine decrypts the AES-protected RSA private key required for signing certificate data. The RSA private key is decrypted only during the certificate signing process and is never stored in plaintext, reducing the risk of unauthorized access.

3.3.7 Blockchain Module

The Blockchain Module manages all interactions with the Ethereum blockchain through smart contracts. Its primary functions include recording certificate hashes

during issuance, retrieving blockchain records for certificate verification, maintaining immutable issuance timestamps, and updating certificate revocation status. The module provides transparent, decentralized, and tamper-resistant verification while separating sensitive credential data from public blockchain storage.

3.3.8 SQL Database

The SQL Database serves as the secure off-chain storage repository for certificate data. It includes storing signed certificate information, retrieving credential data during verification. By storing digitally signed certificate data off-chain, the SQL database preserves data privacy, reduces blockchain storage requirements and transaction costs, and supports efficient credential management.

3.4 Hybrid Storage Architecture

To balance security, privacy, scalability, and storage efficiency, EduCredChain adopts a hybrid storage architecture that combines the immutable properties of blockchain with the flexibility of SQL data storage. Instead of storing complete certificate records on the blockchain, the framework distributes data across two complementary storage layers. Cryptographic proofs are maintained on the blockchain to provide integrity verification, while sensitive certificate information is stored in an SQL database. This approach minimizes blockchain storage requirements while preserving data on a private database and enabling efficient credential management.

3.3.1 On-chain Blockchain Ledger

The blockchain ledger functions as the immutable trust layer of the proposed framework. During certificate issuance, a unique SHA-256 hash is generated from the certificate information and recorded on the Ethereum blockchain through a smart contract. Along with the certificate hash, the ledger stores issuance timestamps and certificate revocation status. Since blockchain records are immutable, any attempt to modify the records are computationally infeasible. By storing only cryptographic proofs rather than complete certificate data, the blockchain provides transparent and decentralized integrity verification without exposing sensitive information.

3.3.2 Off-chain SQL Database

The SQL database serves as the secure and easily accessible storage location for certificate data. Before storage, sensitive certificate information is signed using RSA cryptography to protect it from unauthorized access.

The database maintains certificate identification number, date of issue and signed certificate details including issuance information like name, course, grade etc. Off-chain storage enables efficient querying, updating, and retrieval of credential data while avoiding the storage limitations and transaction costs associated with blockchain networks.

3.3.3 Integration of On-Chain and Off-Chain Storage

The blockchain ledger and SQL database operate collaboratively to provide secure credential management architecture. During certificate issuance, the signed certificate data is stored in the SQL database, while the corresponding cryptographic hash and certificate status are recorded on the blockchain. During verification, the signed certificate information is retrieved from the SQL database and verified with issuers RSA public key. SHA-256 hash is then recomputed and compared with the immutable blockchain record. A matching hash confirms that the certificate has not been altered since issuance. Similarly, certificate revocation requests update the revocation status on the blockchain while maintaining synchronized records within the SQL database, ensuring consistency between both storage layers.

3.3.4 Advantages of the Hybrid Storage Method

The proposed hybrid storage architecture offers several advantages over conventional blockchain-only and centralized storage approaches:

- **Enhanced Data Privacy**, Sensitive certificate information is securely stored in signed off-chain storage, while only cryptographic hashes are maintained on the blockchain.
- **Tamper-Resistant Verification**, Immutable blockchain records enable reliable detection of unauthorized modifications by comparing stored and computed certificate hashes.
- **Reduced Blockchain Storage Overhead**, Storing only cryptographic proofs on-chain significantly minimizes blockchain storage requirements and transaction costs.
- **Improved Scalability**, Off-chain storage prevents blockchain bloat, allowing the system to efficiently manage a large volume of educational credentials.
- **Efficient Data Retrieval**, Certificate data can be quickly accessed and managed through the SQL database without querying the blockchain for complete records.

- **Enhanced Security,** The combination of blockchain integrity, AES-protected key management, and PKI based signed database storage provides multiple layers of security against unauthorized access and data manipulation.
- **Resistance to Database Tampering,** Even if the off-chain database is compromised or manipulated, certificate integrity can still be independently validated using the immutable blockchain records.
- **Transparent Certificate Lifecycle Management,** Certificate issuance, verification, and revocation are securely synchronized between the blockchain and off-chain storage, ensuring consistency and traceability.
- **Cost-Effective Deployment,** The hybrid approach reduces blockchain transaction costs while maintaining the security benefits of decentralized verification.
- **User-Friendly Verification,** The architecture enables web-based certificate verification without requiring users to manage blockchain wallets or possess blockchain-specific technical knowledge.

3.5 Data Flow Overview

The EduCredChain framework manages the complete lifecycle of educational credentials through three primary operations: certificate issuance, certificate verification, and certificate revocation. Each operation integrates cryptographic processing, blockchain interaction, and secure database management to ensure the authenticity, integrity, and validity of digital certificates.

3.4.1 Certificate Issuance

Certificate issuance begins when an authorized issuer submits the certificate through the web interface along with issuer code that acts as an authorization method. The system generates a unique cryptographic hash from the certificate information, which serves as its cryptographic fingerprint. The certificate data (except certificate number and issue data) is then signed using RSA private key and securely stored in the off-chain SQL database along with raw certificate number and SHA-256

hash. Simultaneously, the generated certificate hash, issuance timestamp, and certificate status are recorded on the Ethereum blockchain through a smart contract. This process establishes an immutable proof of authenticity while preserving the contents of certificate data.

3.4.2 Certificate Verification

During verification, the user provides the certificate number through the web interface. The system retrieves the corresponding signed certificate data from the SQL database and verify it using the RSA verification engine with RSA public key. SHA-256 hash is regenerated from the retrieved certificate information and compared with the immutable hash stored on the blockchain. The system also verifies the certificate's revocation status before declaring the credential as valid. If the generated and stored hashes match and the certificate have not been revoked, the certificate is confirmed to be authentic and unaltered.

3.4.3 Certificate Revocation

Certificate revocation is initiated by an authorized issuer when a credential is expired or no longer valid. The system first verifies the issuer's authorization before processing the revocation request. Rather than modifying or deleting the original blockchain record, the smart contract updates this by creating a new blockchain transaction with certificate status as revoked. This transaction is subsequently included in a newly mined block, while the original certificate issuance record remains permanently preserved on the blockchain. As a result, the blockchain maintains a complete and immutable audit trail containing both the issuance and revocation history of the credential. During subsequent verification requests, the system checks both the integrity of the certificate and its current revocation status on the blockchain, ensuring that revoked credentials are immediately identified and rejected without compromising the immutability of previous blockchain records

4. Proposed Methodology

4.1 Certificate Issuance Methodology

4.1.1 Algorithm : Certificate Issuance

Input: Issuer code, Certificate Number, Name, Course, Grade.

Output: Confirmation of Certificate issuance.

1. Start
2. Accepts certificate details as user input.(Certificate Number, Name, Course, Grade)
3. Accepts issuer code as user input and compute its SHA-256 hash.
4. Send an API request To the backend layer with a payload containing issuer code hash and certificate data.
5. Authenticate the issuer using the issuer code hash entered by user.
If authentication fails “Return, Authentication failure” go to 15
6. Generate the SHA-256 hash of the certificate data.
7. Hash issuer code hash using SHA-256 algorithm to serve as AES-256 key.
8. Decrypt the RSA private key using the AES key.
9. Sign the certificate data (excluding the certificate number) using the RSA private key.
10. Store the Certificate number, signed certificate data and certificate hash in the SQL database.
11. Record the certificate hash, issuance timestamp, and initial certificate status on the blockchain ledger.
12. Generate the transaction hash and blockchain record
13. Confirm successful certificate issuance.
14. Return the certificate identifier and transaction details to application layer.
15. End

4.1.2 Flowchart: Certificate Issuance

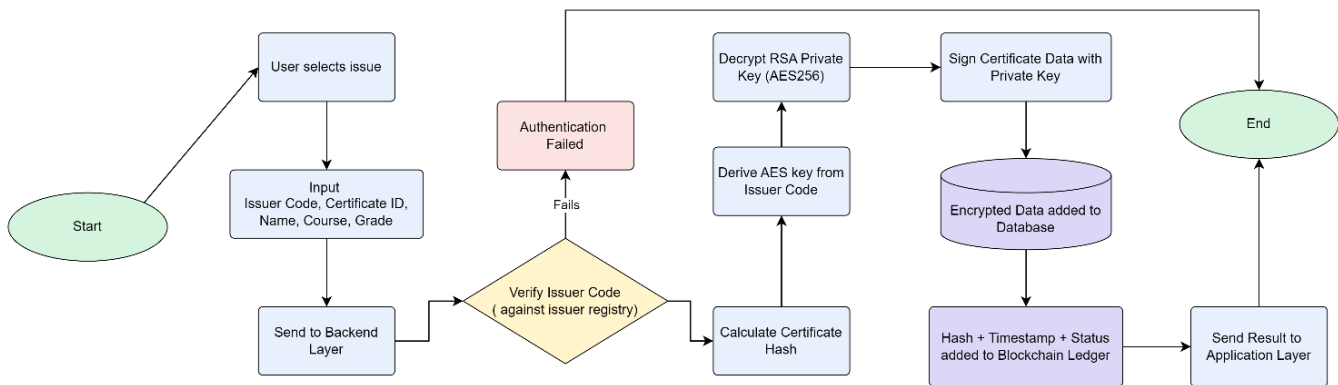


Fig 2, Certificate Issuance workflow

4.2 Certificate Verification Methodology

4.2.1 Algorithm : Certificate Verification

Input: Certificate Number.

Output: Certificate data, Existence status of Certificate in blockchain.

1. Start
2. Accepts certificate number as user input.
3. Send an API request To the backend layer with a payload containing certificate number.
4. Retrieve the certificate record from database with certificate number as primary key.
5. Verify the certificate data using RSA public key.
If verification fails “Return, Integrity Failure” go to 13
6. Recalculate certificate hash with fetched data.
7. Compare the recalculated certificate hash with the stored certificate hash,
If verification fails “Return, Integrity Failure” go to 13
8. Return the certificate data, Certificate Hash to application layer.
9. User confirms certificate details and selects to verify on blockchain.
10. Verify the existence of certificate hash on blockchain along with revocation status.
11. Return the verification result to application layer.
12. End

4.2.2 Flowchart: Certificate Verification

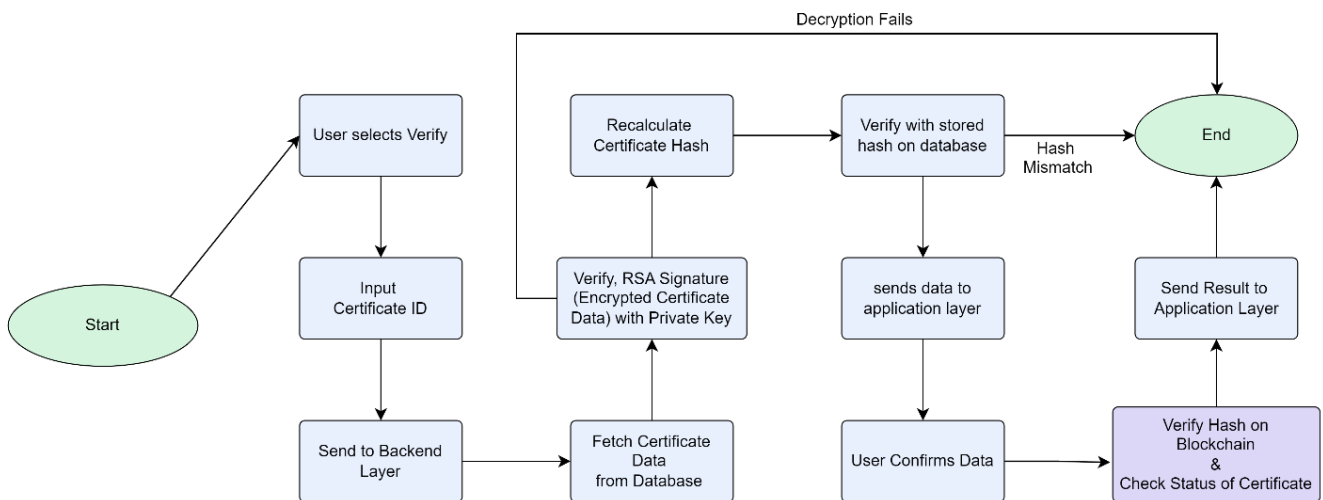


Fig 3, Certificate verification workflow

4.3 Certificate Revocation Methodology

4.3.1 Algorithm : Certificate Revocation

Input: Issuer code, Certificate Number.

Output: Certificate data, Confirmation of Certificate revocation.

1. Start
2. Accepts certificate number as user input.
3. Accepts Issuer Code as user input and compute its SHA-256 hash.
4. Send an API request To the backend layer with a payload containing issuer code hash and certificate number.
5. Authenticate the issuer using the issuer code hash entered by user.
If authentication fails “Return, Authentication failure” go to 15
6. Retrieve the certificate record from database with certificate number as primary key.
7. Verify the certificate data using RSA public key.
If verification fails “Return, Integrity Failure” go to 15
8. Recalculate certificate hash with fetched data.
9. Compare the recalculate certificate hash with hash stored in database
If verification fails “Return, Integrity Failure” go to 15
10. Return the certificate data, Certificate Hash to application layer.
11. User confirms certificate details and initiates revocation on blockchain.
12. Create a new blockchain transaction that updates the certificate status to “Revoked” while preserving the original issuance record.
13. Generate the transaction hash and blockchain record.
14. Return the certificate identifier and transaction details to application layer.
15. End

4.3.2 Flowchart: Certificate Revocation

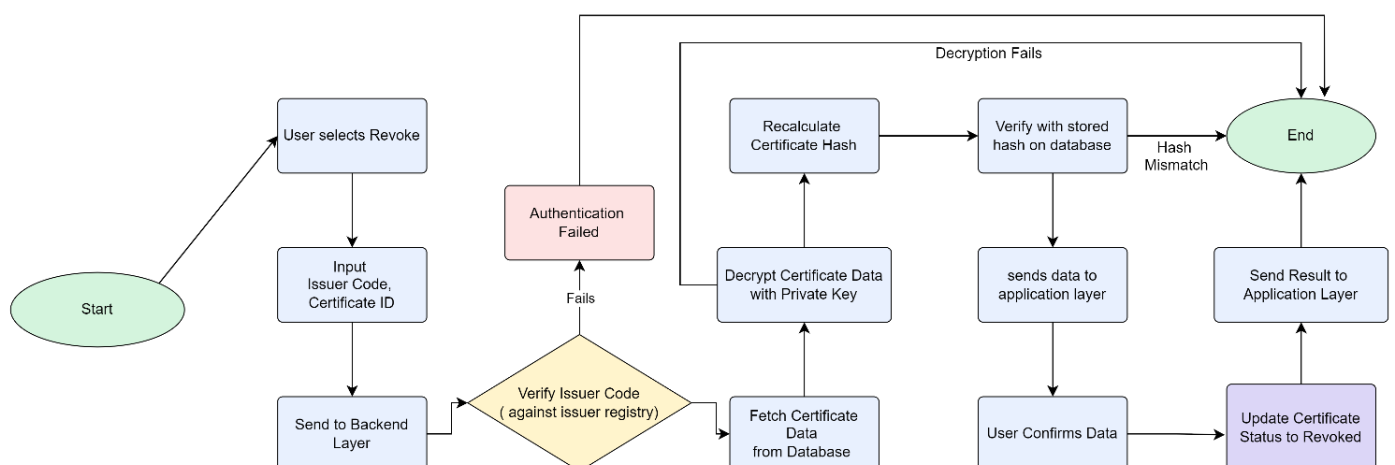


Fig 4, Certificate Revocation workflow

4.4 PKI-Based Signature and Key Management

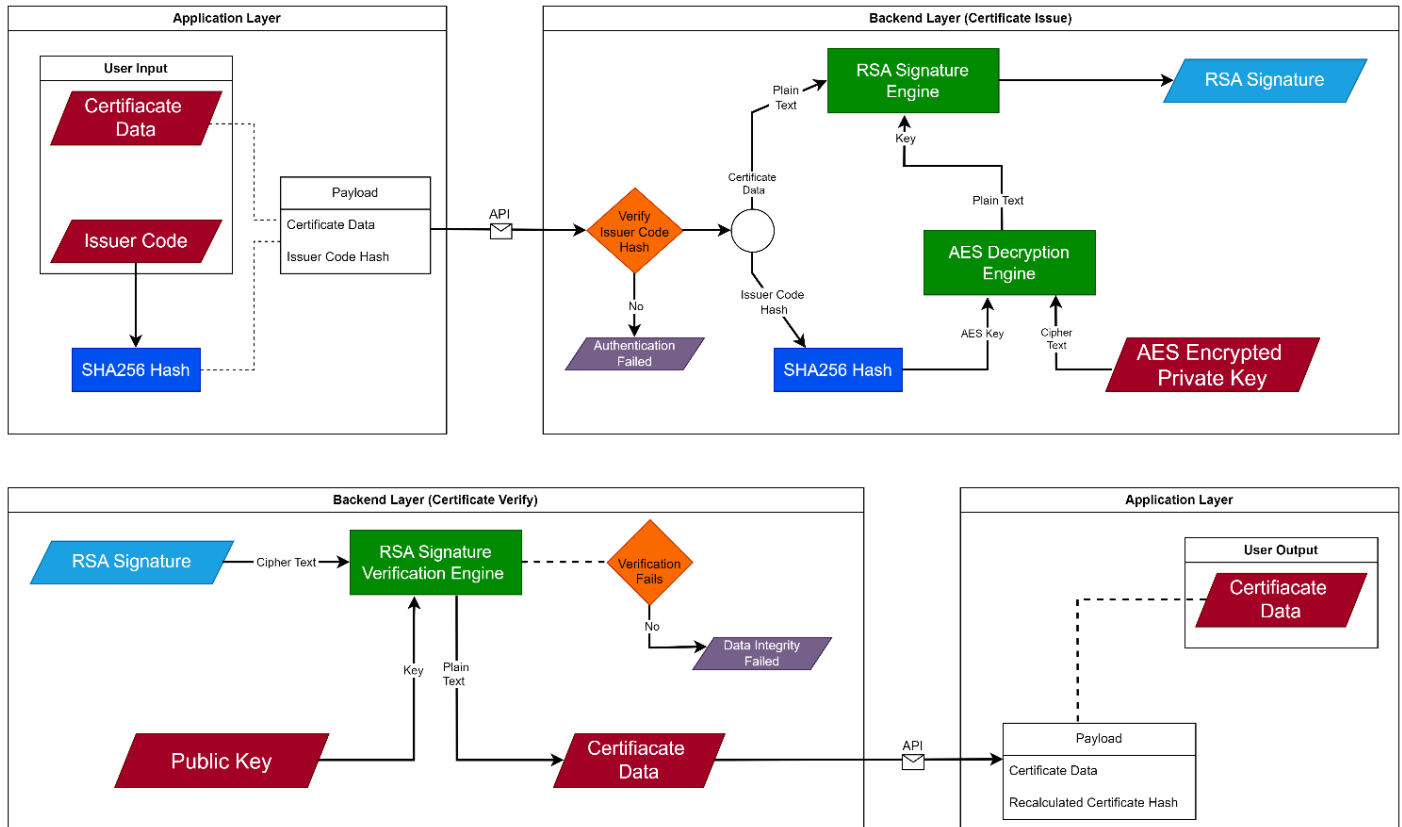


Fig 5, Architecture diagram PKI infrastructure

Figure 3.2.1 illustrates the PKI infrastructure certificate issuance and verification workflow adopted in the proposed EduCredChain system as explained in algorithms 4.1.1 and 4.2.1. During certificate issuance, the issuer submits the certificate information along with an issuer code through the application layer. The issuer code is hashed using the SHA-256 algorithm and transmitted with the certificate data to the backend for authentication. Upon successful authentication, the RSA private key is recovered by decrypting the AES-256 encrypted private key using AES key derived from the issuer code hash. The recovered private key is then utilized by the RSA Signature engine to sign the

certificate data before storage (clearly demonstrated in fig 4.1.2). During certificate verification, the signed certificate data is retrieved and verified by the RSA verification engine using the corresponding RSA public key to recover the original certificate information. If the verification process is successful, the verified certificate data is returned to the application layer. Otherwise, the certificate is identified as invalid and the verification request is rejected (clearly demonstrated in fig 4.2.2). This workflow ensures that certificate issuance is restricted to authenticated issuers while enabling secure recovery and verification of credential data through a hybrid AES–RSA PKI mechanism.

4.5 Certificate Hash Generation

4.5.1 Algorithm : Certificate Hash Generation Function

Input: Certificate details (Certificate ID, Student Name, Course ID, Grade, Issue Date)

Output: SHA-256 certificate hash.

1. Start
2. Receive the certificate details.
3. Normalize the input fields:
 - a. Convert Student Name to lowercase and remove all whitespaces.
 - b. Convert Course ID to lowercase and remove all whitespaces.
 - c. Convert Certificate ID to lowercase and remove all whitespaces.

- d. Convert the grade to lowercase (if alphanumeric) and remove all whitespace characters. If the grade is numeric, preserve its numeric representation.
- e. Preserve the Issue Date in its standard DDMMYYYY format.
4. Concatenate the normalized certificate fields into a single string in the following order: Certificate ID, Student Name, Course ID, Issue Date, Grade, Salt (In real time Implementation) .
5. Apply the SHA-256 hashing algorithm to the concatenated string to generate the 256-bit unique certificate hash.
6. Return the generated hash.
7. End

4.5.2 Architecture of Hash Generation Module

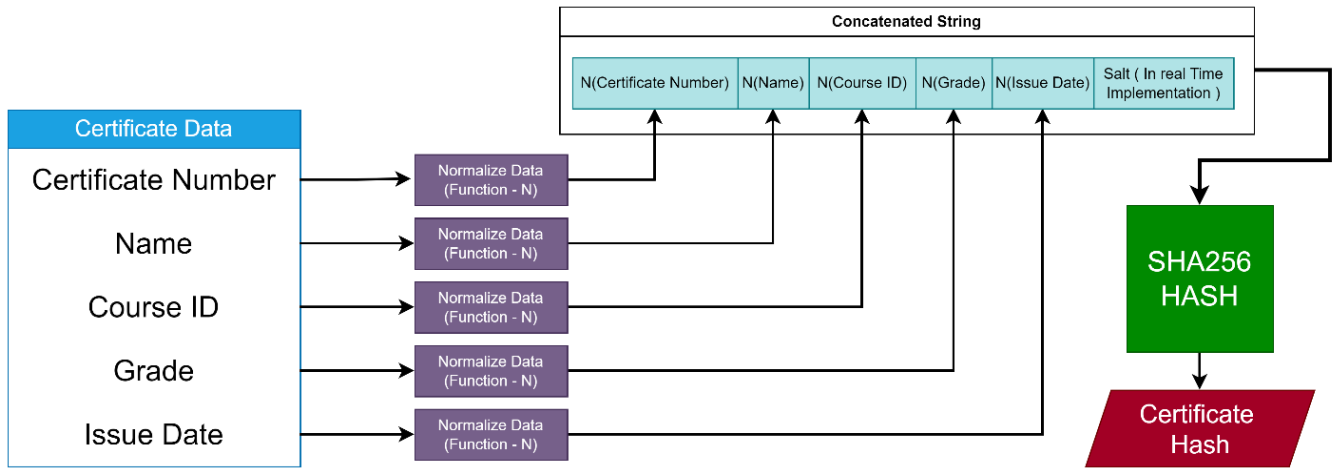


Fig 6, Architecture diagram, Hash Generation Module

4.5.3 Mathematical Representation of Hash Generation Function

$$H = SHA256 (N(cn) || N(n) || N(cid) || N(g) || N(d) || S)$$

Where,

H = Generated certificate hash

cn = Certificate Number

n = Name

cid = Course ID

g = Grade

d = Issue Date

S = Optional runtime salt (used only in production deployments)

|| = Concatenation operator

$N(x)$ = Normalization function that converts x to lowercase and removes all whitespace in x.

7. Blockchain Registration

The generated SHA-256 certificate hash is stored on the blockchain as a unique cryptographic proof of integrity. Since only the hash is recorded on-chain rather than the complete certificate data, the system preserves data privacy while enabling transparent and reliable

integrity verification during certificate validation. The backend invokes an Ethereum smart contract to register the certificate on the blockchain. The smart contract validates the request, records the certificate hash, and executes the transaction, ensuring that the credential is securely and immutably registered on the blockchain. During certificate registration, the blockchain

automatically records the transaction timestamp, indicating the exact certificate issuance date and time to provide a verifiable record of issuance and supports chronological tracking and auditing of certificate transactions. Upon successful certificate issuance, the smart contract initializes the certificate status as “Valid” (or Active). If the certificate is revoked in the future, the status is updated through a new blockchain transaction while preserving the original issuance record.

4.6.1 Real-World Deployment of EduCredChain

The proposed EduCredChain framework is designed to operate on a permissioned blockchain network managed by a university. During the initial deployment phase, the university establishes the blockchain network and serves as the issuing authority responsible for deploying the smart contract. The deployment process is performed manually by an authorized blockchain administrator to ensure that the contract owner address is securely initialized and cannot be impersonated or maliciously replaced. Once deployed, the blockchain permanently records the university’s Ethereum address as the contract owner. Each affiliated college operates as a blockchain node that maintains a

synchronized ledger, validates transactions, and participates in consensus. Although all nodes maintain the ledger, only the registered contract owner can issue or revoke certificates. When a certificate is issued, the university’s backend system generates the certificate hash and submits it to the deployed smart contract using the owner’s blockchain account. The transaction is validated by participating nodes and recorded in a new block. After consensus, the updated ledger is synchronized across all nodes, creating an immutable record of issuance. Similarly, certificate revocation follows the same workflow. Certificate revocation follows the same process, with the owner submitting a revocation transaction that is validated and recorded on the blockchain while preserving the original issuance record. The manual initialization of the blockchain network and smart contract deployment serves as the root of trust for the entire framework. Since the contract owner is established only once during deployment and all privileged operations require transactions from the registered owner address, Consequently, unauthorized users or malicious nodes cannot impersonate the issuing authority or create fraudulent certificate records, even if they participate in the network.

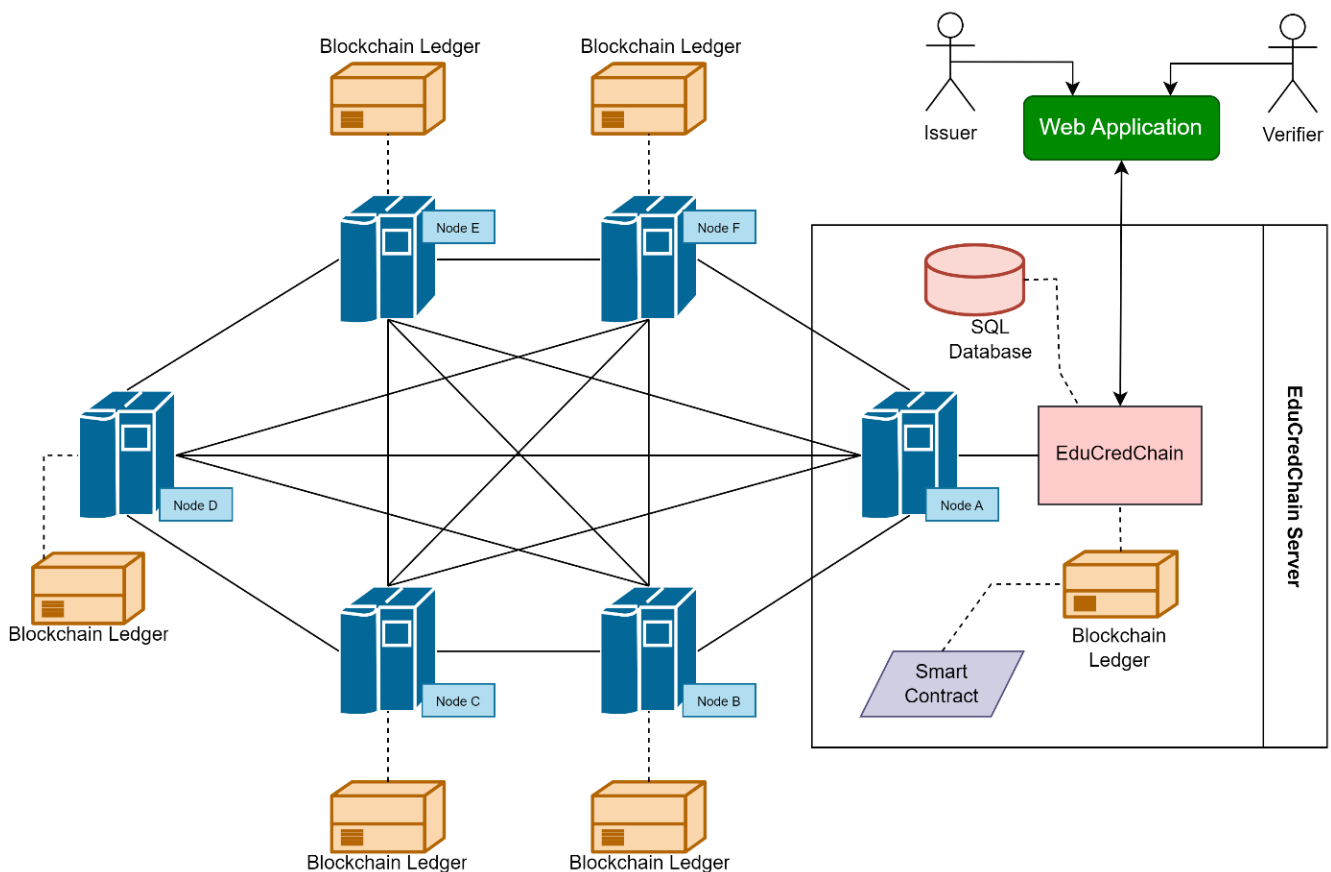


Fig 7, Permissioned Ethereum Blockchain Network (Real time deployment model)

4.6.2 Algorithm: Certificate Registration in Smart Contract

Input: Certificate Hash

Output: Certificate successfully registered on the blockchain

1. Start
2. Receive the certificate hash.
3. Verify that the caller is the contract owner.
4. Check whether the certificate hash already exists.
If existing Return "Duplicate Certificate", go to 12.
5. Create a new certificate record with certificate hash.
6. Set the issuance timestamp to the current blockchain timestamp.
7. Set the certificate existence status to True.
8. Set the revocation status to False.
9. Set the revocation timestamp to 0.
10. Store the certificate record in the smart contract state.
11. Emit the CertificateIssued event.
12. End

4.6.3 Algorithm: Certificate Verification in Smart Contract

Input: Certificate Hash

Output: Certificate status, revocation status, issuance timestamp, and revocation timestamp

1. Start
2. Receive the certificate hash.
3. Search for the corresponding certificate record.
4. Retrieve the stored certificate record.
If not found Return, "certificate not found" go to 6.
5. Return: Certificate existence status, Revocation status, Issuance timestamp, Revocation timestamp
6. End

4.6.4 Algorithm: Certificate Revocation in Smart Contract

Input: Certificate Hash

Output: Certificate revocation recorded on the blockchain

1. Start
2. Receive the certificate hash.
3. Verify that the caller is the contract owner.
4. Check whether the certificate exists.
If not found "Return, certificate not found" go to 10.
5. Check whether the certificate is already revoked.
If already revoked, If not found Return, "Already Revoked" go to 10.
6. Update the certificate revocation status to True.
7. Record the current blockchain timestamp as the revocation timestamp.
8. Update the certificate record in the smart contract state. The resulting transaction is subsequently recorded in a new blockchain block.
9. Emit the Certificate Revoked event.
10. End

4.6.5 Algorithm: Owner Authentication

Input: Transaction sender address (msg. sender)

Output: Authorization status

1. Start
2. Receive the transaction request.
3. Retrieve the sender's blockchain address (msg. sender).
4. Retrieve the stored contract owner address.

5. Compare the sender address with the owner address.
6. If both addresses match: Grant authorization and Continue smart contract execution.
Otherwise: Reject the transaction. Return “Access Denied” go to 7.
7. End

5. Implementation

5.1 Development Environment

The EduCredChain prototype was implemented using a combination of blockchain technologies, web development frameworks, cryptographic libraries, and database management tools. Python was used as the primary programming language for implementing the backend application logic, while FastAPI served as the RESTful web framework for API communication. The user interface was developed using HTML, CSS, and JavaScript to provide an interactive web-based platform. A local Ethereum blockchain was established using Ganache, and the smart contract was implemented in

Solidity. Communication between the backend and the blockchain was achieved using the Web3.py library. The Cryptography library was employed to implement RSA encryption/decryption for signature and AES-based key protection, whereas hashlib was utilized for generating SHA-256 certificate hashes. Certificate data was securely stored in an SQLite database as part of the hybrid storage architecture. The application was developed using Visual Studio Code, managed through Git, and maintained using a GitHub repository (<https://github.com/Abhiram-ARS/EduCredChain>) for version control and collaborative development. The complete development environment is summarized in Table 5.2.

5.2 Technology Stack

Technology	Category	Application
Python	Programming Language	Implements the backend application logic, cryptographic operations, and system integration.
HTML, CSS, JavaScript	Frontend Technologies	Develops the responsive web-based user interface for certificate issuance, verification, and revocation, enabling user interaction with the backend services.
FastAPI	Web Framework	Provides RESTful APIs for certificate issuance, verification, and revocation while facilitating communication between the frontend, blockchain, and database.
SQLite	Database Management System	Stores signed certificate data, certificate identifiers, timestamps, and other off-chain information.
Ganache	Local Ethereum Blockchain	Provides a local Ethereum test network for smart contract deployment, transaction processing, and prototype evaluation.
Solidity	Smart Contract Language	Implements blockchain-based certificate registration, verification, and revocation through Ethereum smart contracts.
Web3.py	Blockchain Interface Library	Enables interaction between the Python backend and Ethereum blockchain, including smart contract deployment and transaction execution.
Cryptography	Python Cryptographic Library	Provides RSA encryption/decryption to implement digital signature generation and verification, and AES-based key management for securing certificate data and cryptographic keys.

Hashlib	Python Hashing Library	Implements the SHA-256 hashing algorithm for generating unique cryptographic hashes used in certificate integrity verification.
Visual Studio Code	Integrated Development Environment (IDE)	Used for application development, smart contract programming, debugging, and project management.
GitHub	Version Control and Code Repository	Maintains source code, version control, collaborative development, and project documentation throughout the software development lifecycle.

Table 3, Technology Stack Used

5.3 Backend Implementation

The backend of the proposed EduCredChain framework was implemented using Python and the FastAPI web framework to provide a secure and modular software layer between the user interface, cryptographic services, SQL database, and Ethereum blockchain. Communication between the frontend and backend is achieved through RESTful APIs, where certificate issuance, verification, and revocation requests are transmitted using JSON-formatted payloads. FastAPI performs request validation before invoking the corresponding backend services.

The backend integrates multiple functional modules responsible for issuer authentication, SHA-256 hash generation, RSA digital signature generation and verification, AES-based private key recovery, database

management, and blockchain interaction. Database operations are performed on the off-chain SQL database for storing and retrieving certificate records, while blockchain transactions are executed through the Web3.py library to invoke the deployed Ethereum smart contract. The backend records only the certificate hash and certificate status on the blockchain, whereas signed certificate information is maintained in the SQL database as part of the hybrid storage architecture.

This implementation follows a modular software architecture that separates the presentation layer, business logic, cryptographic processing, database management, and blockchain services. Such separation improves maintainability, scalability, and interoperability while enabling secure coordination among all system components during certificate lifecycle operations.

5.4 Smart Contract Structure

5.4.1 Smart contract design and structure

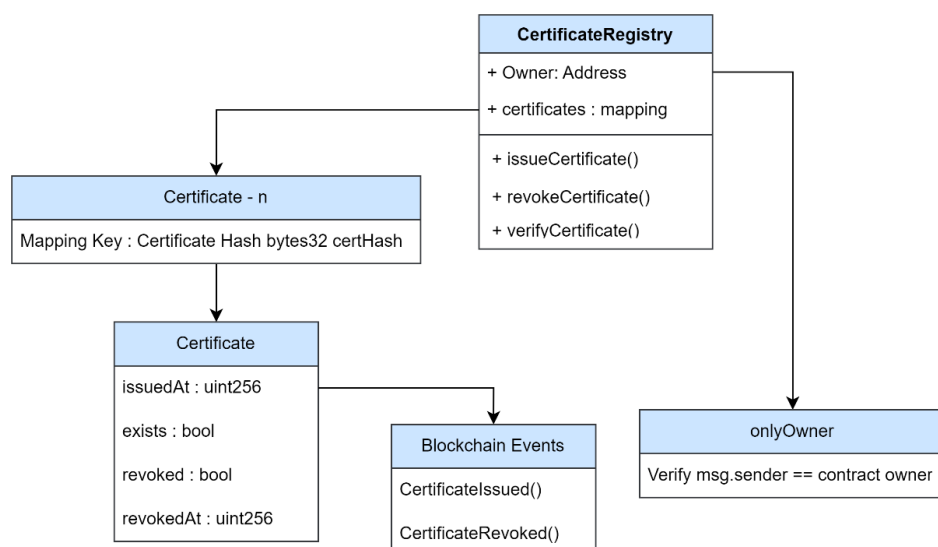


Fig 8, Smart contract design and structure

5.4.2 Gas optimization

The proposed EduCredChain framework minimizes blockchain resource consumption through a hybrid storage architecture. Instead of storing complete certificate data on the blockchain, only the SHA-256 certificate hash, issuance timestamp, and certificate status are recorded on-chain, while the digitally signed certificate information is maintained in the off-chain SQL database. This significantly reduces transaction data size, blockchain storage requirements, and gas consumption during certificate issuance and revocation. Furthermore, certificate verification utilizes read-only (view) smart contract functions, which execute locally without generating blockchain transactions, thereby incurring zero gas cost. This design achieves efficient blockchain utilization while preserving certificate integrity, authenticity, and transparent verification.

5.4.3 Access Control Mechanism in Smart Contract

The EduCredChain smart contract employs an owner-based access control mechanism to prevent unauthorized users to issue or revoke certificates. During smart contract deployment, the blockchain address of the deploying authority is automatically registered as the contract owner. Privileged functions such as certificate issuance and revocation are protected using the `onlyOwner` constrain, which verifies that the transaction originates from the original owner before execution. Any unauthorized transaction is immediately rejected, while certificate verification remains publicly accessible to

ensure transparent and independent validation of educational credentials.

5.4.4 Event Logging

The smart contract generates blockchain events to provide a transparent and immutable audit trail of certificate lifecycle operations. Upon successful certificate issuance, the `CertificateIssued` event records the certificate hash and issuance timestamp. Similarly, the `CertificateRevoked` event is triggered when a certificate is revoked, storing the certificate hash and revocation timestamp. These events enable efficient transaction monitoring, improve traceability, and allow external applications to track certificate status changes without directly querying the blockchain storage.

5.5 Database Design

EduCredChain employs an SQLite database (on testing environment) as the off-chain storage component of the hybrid Blockchain–PKI architecture. The database stores certificate metadata required for credential management, while the corresponding SHA-256 certificate hash is also maintained to facilitate integrity verification. During certificate verification, the retrieved certificate information is processed to generate a new SHA-256 hash, which is compared against the immutable blockchain record to confirm certificate authenticity. This hybrid storage approach enables efficient data retrieval while reducing blockchain storage overhead and preserving the integrity of educational credentials.

Field Name	Data Type	Sign Status	Description
Certificate_No*	<i>VARCHAR(11)</i>	Not Signed	Unique certificate identifier (Primary Key).
Name	<i>VARCHAR(30)</i>	Signed	Name of the certificate holder.
Course_ID	<i>VARCHAR(5)</i>	Signed	Unique identifier of the course or program.
Issue_Date	<i>DATE</i>	Signed	Date on which the certificate was issued.
Grade	<i>INTEGER</i>	Signed	Grade or academic score obtained by the student.
Hash	<i>VARCHAR(64)</i>	Not Signed	SHA-256 hash of the certificate used for blockchain integrity verification.
* Primary Key			

Table 4, SQL Database Architecture

5.6 User Interface

5.6.1 Issue page

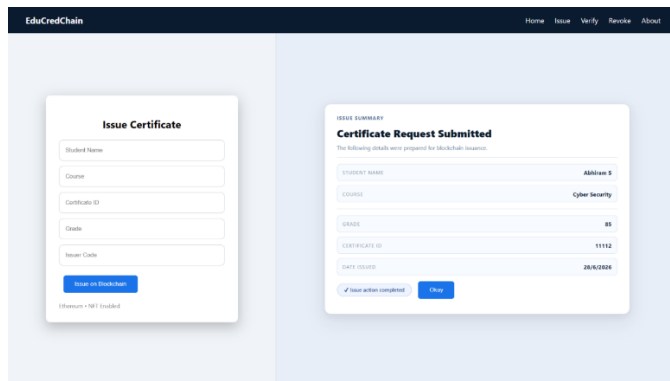


Fig 10, Certificate Issuance Interface

The certificate issuance interface enables authorized issuers to enter certificate details and authenticate using the issuer code. Upon successful authentication, the backend generates the certificate hash, digitally signs the certificate, stores the signed data in the off-chain database, and registers the certificate hash on the blockchain.

5.6.2 Verify page

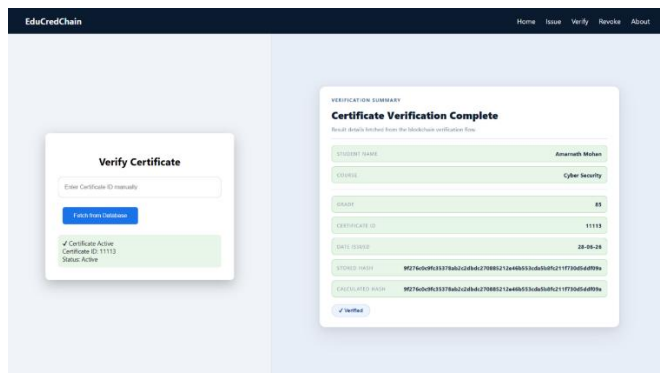


Fig 11, Verification of a Valid Certificate

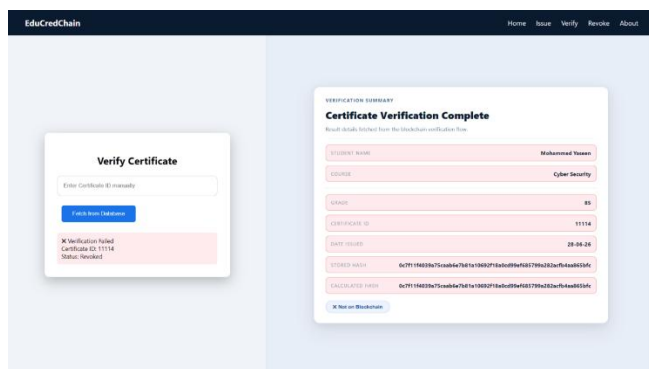


Fig 12, Verification of a Revoked Certificate

The verification interface allows users to validate educational credentials by entering the certificate number.

The system retrieves the signed certificate data, performs RSA signature verification, regenerates the SHA-256 hash, compares it with the blockchain record, and displays the certificate status. Figures 5.4.3-1 and 5.4.3-2 illustrate successful verification of valid and revoked certificates, respectively.

5.6.4 Revoke page

The revocation interface enables authorized issuers to revoke previously issued certificates after successful authentication. The backend verifies the certificate integrity before submitting a blockchain transaction that updates the certificate status while preserving the original issuance record.

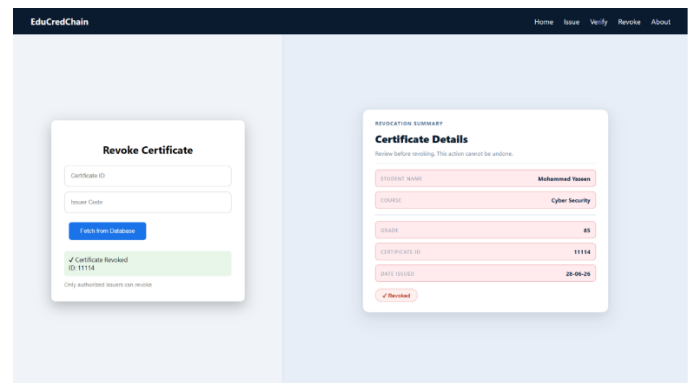


Fig 13, Certificate Revocation Interface

6. Experimental Evaluation

6.1 Hardware and software Configuration of testing environment

- Intel Core i5-12450H
- 16 GB RAM
- Windows 11
- Ganache v7.x

6.2 Evaluation Methodology

6.2.1 Functional Evaluation

The functional evaluation was conducted to verify the correctness of the proposed EduCredChain framework across the complete certificate lifecycle. The evaluation validated certificate issuance, verification, and revocation by ensuring that certificate hashes were correctly generated, securely recorded on the blockchain, and synchronized with the off-chain SQL database. The functional evaluation was conducted to verify whether valid certificates were successfully authenticated,

revoked certificates were accurately identified, and certificate lifecycle operations behaved as proposed.

6.2.2 Security Evaluation

The security evaluation assessed the framework's ability to protect educational credentials against common security threats, including certificate forgery, unauthorized certificate issuance, and database tampering. A dedicated database tampering simulation module was developed to emulate unauthorized modifications to the off-chain SQL database. The evaluation examined whether modifications to certificate data resulted in a mismatch between the recomputed hash and the immutable blockchain record, enabling the detection of compromised credentials. Furthermore, the owner-based access control mechanism successfully prevented unauthorized users from issuing or revoking certificates.

6.3 Results

6.3.1 Functional Evaluation

Function	Expected Outcome	Result	Observation
Initial Setup	Deployment of smart contract	Pass	New block is mined.
	Creation of Table in database	Pass	File created with required fields
Certificate Issuance	Hash stored on blockchain	Pass	New block is mined.
	RSA private key recovery	Pass	Key derived from issuer code.
	Sign certificate data	Pass	Database stores ciphertext.
	Data stored on database	Pass	New record is inserted.
Certificate Verification	Record Retrieval from database	Pass	Retrieves record from database.
	RSA signature verification	Pass	Successfully recovered the original certificate data.
	Recalculate same hash with data	Pass	Hash matched while processing.
	Blockchain hash verification	Pass	Generated hash matched the immutable blockchain record.
Certificate Revocation	Revoked status updated	Pass	New block is mined.
	Verify Revoked Certificate	Pass	Shows certificate is revoked
System Integration	API communication	Pass	Successful communication between system components
	Layer integration	Pass	Successful integration of all system components
	Issuer code authentication	Pass	Fails with incorrect input

Table 5, Result of functional evaluation of EduCredChain

The functional evaluation results in Table 6.2.1 demonstrate the operational feasibility of the proposed EduCredChain framework across the certificate lifecycle. During setup, the smart contract was deployed and the required database tables were created. Certificate issuance successfully generated and recorded the

6.2.3 Performance Evaluation

The performance evaluation measured the operational efficiency of the proposed framework by analysing the execution time and blockchain gas consumption associated with certificate issuance, verification, and revocation. Each experiment was conducted by Each experiment was performed using a batch of 50 certificates processed sequentially, and the reported execution times represent the average observed latency. The evaluation also examined the impact of the hybrid storage architecture on certificate processing and retrieval. The performance evaluation examines whether the proposed framework achieves efficient certificate management while maintaining secure blockchain registration, rapid verification, and low blockchain storage overhead.

certificate hash on the blockchain, securely derived the RSA private key from the issuer code, generated RSA signatures, and stored the signed records in the SQL database. Certificate verification successfully validated the stored RSA digital signatures, regenerated the SHA-256 hash, and confirmed its integrity against the

blockchain record. Certificate revocation correctly updated the blockchain status, enabling revoked certificates to be identified during verification. successful API communication, seamless interaction between the

application, blockchain, and database layers, and effective issuer code authentication further confirm the reliability of the proposed hybrid Blockchain–PKI architecture.

6.3.2 Performance Evaluation

6.3.2.1 Gas Consumption Analysis

Operation	Gas Used	Said et al. [1]	Serranito [45]
Smart Contract Deployment	549,244	Not Recorded	Not Recorded
Certificate Issuance	73,007	418,888	Not Recorded
Certificate Revocation	53,667	260,705	Not Recorded
Certificate Verification	0	0	Not Recorded

Table 6, Comparative gas analysis with similar papers for different functions

Certificate verification invokes only read-only smart contract functions (view functions), which are executed locally by Ethereum nodes without generating blockchain transactions. Therefore no gas is consumed.

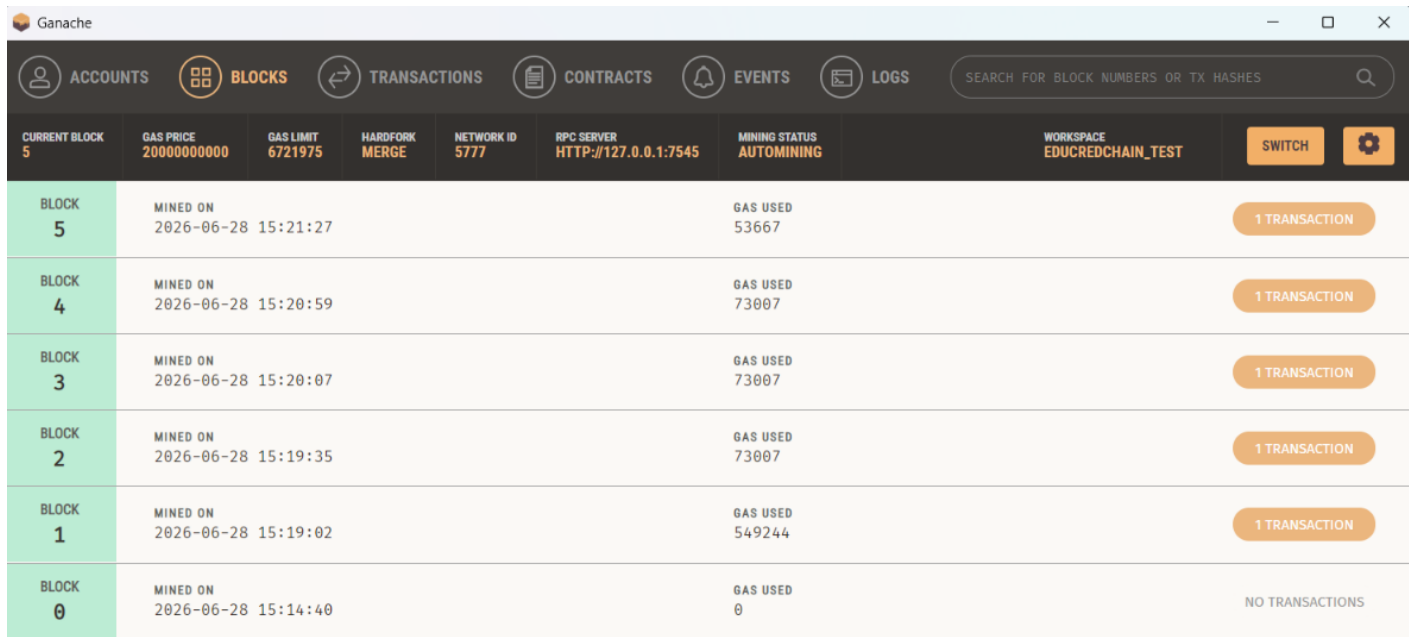


Fig 14, Blockchain record in ganache. Block 1: Smart Contract Deployment. Block 2-4: Certificate Issuance. Block 5: Certificate Revocation

The gas consumption analysis in Table 6.2.2.1 demonstrates the blockchain resource efficiency of EduCredChain framework. Smart contract deployment required 549,244 gas as a one-time initialization cost. Certificate issuance consumed 73,007 gas, substantially lower than the 418,888 gas reported by Said et al. [1], because only certificate hashes and status information are stored on-chain. Similarly, Certificate revocation required 53,667 gas, compared to 260,705 gas in Said et al. [1], demonstrating significantly lower transaction costs.

Certificate verification consumed no gas because it uses a read-only (view) smart contract function that does not create a blockchain transaction. The blockchain records shown in Figure 6.2.2.1 further confirm that certificate issuance and revocation generate new blocks, while verification performs only a blockchain query. Overall, the results indicate that EduCredChain achieves secure credential management with considerably lower blockchain resource utilization than comparable blockchain-based credential verification frameworks.

6.3.2.2 Performance Evaluation

Work	Test Environment	Average Issue Latency (s)	Average Revoke Latency (s)	Average Verify Latency (s)
ElimuChain[1]	BNB Smart Chain test deployment	12.000	11.540	3.830
Serranito [45]	Ethereum/private blockchain	38.206	38.627	0.915
EduCredChain	Ganache + SQLite (local)	0.123*	0.126*	0.012*
* These values should be interpreted as relative implementation performance rather than direct blockchain performance.				

Table 7, Performance evaluation with related papers

Note: Latency measurements for EduCredChain were obtained in a local Ganache-based environment and primarily represent application processing time, including cryptographic operations, database access, smart contract invocation, and transaction execution under immediate block confirmation. In production blockchain networks, additional latency due to network propagation and consensus mechanisms would contribute to the total end-to-end transaction time. Therefore, the reported values should be interpreted as implementation performance under the specified experimental setup rather than direct estimates of public blockchain deployment latency.

The performance comparison in Table 6.3.2.2 shows that EduCredChain achieved average latencies of 0.123 s for certificate issuance, 0.126 s for certificate revocation, and 0.012 s for certificate verification under the Ganache-based test environment with SQLite. These values are considerably lower than those reported for ElimuChain [1] and Serranito [45]. However, these measurements were obtained in a local private blockchain with immediate block confirmation and therefore reflect implementation performance rather than public blockchain latency. The results indicate that the proposed hybrid architecture provides efficient certificate processing by minimizing blockchain overhead through off-chain storage and on-chain hash registration. The comparison highlights the operational efficiency of the implemented prototype under the experimental setup, while acknowledging that end-to-end latency will vary in real-world blockchain deployments depending on network conditions and consensus mechanisms.

6.3.3 Security Evaluation

Assumptions, Attacker can - modify SQL, intercept API, Submit fake certificates, while Attacker cannot - break SHA256, compromise blockchain consensus, obtain owner's Ethereum key

Attack / Scenario	Target	Result	Defensive Mechanism
Simple Database Tampering	Database	Detected	Integrity validation failed.
Database tampering after compromise of RSA private key	Database + PKI	Detected	Stored hash mismatch while initial verification
Database tampering with forged hash replacement	Database + PKI + Hash function	Detected	Block not found on blockchain
Man-in-the-middle interception of issuer code	Network	Failed	Issuer code is hashed within the application layer before transmission.

Unauthorized certificate issuance or revocation from a different blockchain account	Blockchain	Blocked	‘onlyOwner’ constrain in smart contract
Replay Attack	Network	Failed	Previously captured issuance requests cannot alter existing certificates because duplicate hashes are rejected by the smart contract.
Direct blockchain modification attempt	Network	Computationally Impractical	Computationally infeasible due to blockchain immutability and consensus.
Certificate Forgery	Application	Failed	Forged certificate produces a different hash and fails blockchain verification.
Hash Collision Attack	Cryptography	Computationally Impractical	Highly impractical due to the collision resistance of SHA-256.

Table 8, Potential security threat and results

The security evaluation in Table 6.3.3 demonstrates the resilience of EduCredChain against attacks targeting the application, database, cryptographic, network, and blockchain layers. Database tampering was consistently detected through blockchain integrity verification, even if an attacker possessed the RSA private key or modified the stored certificate hash. Unauthorized certificate issuance and revocation from external accounts were prevented by the smart contract's onlyOwner access control. Replay attacks were mitigated by rejecting duplicate certificate hashes within the smart contract, while forged certificates failed verification due to SHA-256 hash mismatches with the immutable blockchain record. Direct modification of blockchain data and SHA-256 hash collision attacks remain computationally impractical under the security assumptions of Ethereum and modern cryptographic hash functions. Overall, the evaluation confirms that the hybrid Blockchain-PKI architecture provides strong protection through signed off-chain storage, blockchain immutability, cryptographic integrity verification, and smart contract authorization.

7. Discussions

Most existing blockchain-based educational credential systems do not report detailed Ethereum transaction gas consumption or performance, limiting direct quantitative comparison. Accordingly, this work reports detailed gas consumption for certificate issuance and revocation to improve reproducibility and facilitate future comparisons.

Functional evaluation confirms the successful implementation of EduCredChain with certificate issuance, verification, revocation, and seamless integration between system layer. The system maintains data integrity and authenticated valid certificates throughout the certificate lifecycle. The performance

evaluation demonstrates efficient operation of the hybrid architecture. Certificate verification achieved the lowest execution time (Table 6.3.2.2) because it executed a read-only blockchain query without transaction creation. Certificate issuance and revocation (fig 6.3.2.2) incurred moderate overhead due to transaction processing. Compared with Said et al. [1], gas consumption was reduced by approximately 79–82% (Table 6.3.2.1) because only certificate hashes are stored on-chain. The security evaluation confirms that the framework effectively detects database tampering, certificate forgery, and unauthorized modifications through blockchain hash verification and owner-based access control. Replay attacks and unauthorized certificate issuance were prevented, while blockchain tampering and SHA-256 collision attacks remained computationally infeasible, demonstrating the security of the hybrid Blockchain-PKI architecture. These results validate the proposed hybrid storage architecture, demonstrating that storing only SHA-256 hashes on-chain significantly reduces blockchain resource consumption without compromising credential integrity or authentication.

8. Limitations

Although the proposed EduCredChain framework demonstrates secure and efficient educational credential management, several limitations remain.

- Private blockchain deployment, The prototype has been evaluated with a private Ethereum (Ganache) network. performance and security characteristics may subject to differ in consortium or public blockchain environments while real time implementation.
- Limited scalability evaluation, The system has been tested on a limited number of certificates, and large-

scale deployment involving millions of credentials has not yet been evaluated.

- Institution-level deployment, The current implementation assumes a single university as the certificate issuing authority and does not address multi-university governance or cross-institution interoperability.
- Dependence on off-chain database availability, While blockchain preserves certificate integrity, retrieval of complete certificate information depends on the availability of the off-chain SQL database.
- Centralized key management, RSA private keys are managed by the issuing institution. Although protected using AES encryption, secure key backup and recovery mechanisms are beyond the scope of this work.

9. Future Work

The proposed EduCredChain framework can be extended in several directions to improve its scalability and practical adoption.

- Deployment on a consortium blockchain for multi-organization participation and advance decentralized governance.
- Development of an efficient and comfortable user experience while verification, and management with mobile application and QR code-based certificate verification.
- Support for multi-institution credential management across universities, colleges, and certification providers under a unified platform.
- Integration with Decentralized Identity (DID) and Verifiable Credentials (VCs) for user-controlled digital identities.
- Evaluation on public blockchain platforms with large-scale scalability testing to understand real time efficiency to handle thousands to millions of certificates and concurrent users simultaneously under real-world network conditions..
- Adoption of Inter Planetary File System (IPFS) or other decentralized storage solutions for distributed off-chain data management.
- Implementation of role-based access control (RBAC) to support multiple administrative roles within institutions.
- Incorporation of post-quantum cryptographic algorithms to improve long-term cryptographic resilience.
- Addition of analytics and audit dashboards for monitoring certificate issuance, verification, and revocation activities.

9.1 Scope Beyond Educational Credential Validation

Although validated using educational credentials, the proposed hybrid Blockchain–PKI architecture is sufficiently generic to support other secure digital document verification applications. It can be applied to other domains requiring secure document issuance, integrity verification, and lifecycle management. The framework can be adapted for professional certifications, digital identity documents, healthcare records, government-issued licenses, legal agreements, and corporate training certificates. By combining blockchain immutability with encrypted off-chain storage, the architecture provides a flexible and scalable approach for managing sensitive digital records while preserving confidentiality, authenticity, and data integrity. This demonstrates that the proposed framework has the potential to serve as a generic secure document verification platform beyond the Educational credential ecosystem.

10. Conclusion

This paper presented EduCredChain, a hybrid Blockchain–PKI framework for secure educational credential management. The framework combines Ethereum, SHA-256 hashing, RSA signatures, AES-based key protection, and hybrid on-chain/off-chain storage to provide secure, transparent, and tamper-resistant certificate management. By storing only certificate hashes on the blockchain and maintaining signed certificate metadata in an off-chain SQL database, the framework achieves both data confidentiality and blockchain storage efficiency. Experimental evaluation demonstrated correct certificate issuance, verification, and revocation, along with low execution times and reduced blockchain gas consumption. Security analysis confirmed the framework's ability to detect database tampering, prevent unauthorized certificate management, and resist certificate forgery, replay attacks, and unauthorized blockchain transactions through blockchain integrity verification and owner-based access control. The primary contribution of this work is a practical hybrid Blockchain–PKI architecture that addresses the limitations of existing blockchain-based credential verification systems by integrating signed off-chain storage, secure key management, and efficient certificate lifecycle management. Future work includes consortium blockchain deployment, multi-institution integration, QR code-based verification, mobile support, and decentralized identity integration to improve scalability and interoperability.

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